

DP014

Physics 1

Semester I

Session 2024/2025

2 hours



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KEMENTERIAN PENDIDIKAN MALAYSIA

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI (PRA-PSPM 1)
PRE-MATRICULATION PROGRAMME EXAMINATION (PRE-PSPM 1)

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

INSTRUCTIONS TO CANDIDATE:

The question paper consists of 10 questions.

Answer all questions.

The use of an electronic calculator is permitted.

QUESTION	MARK(S)
1	/ 8
2	/ 12
3	/ 9
4	/ 12
5	/ 9
6	/ 12
7	/ 18
<i>TOTAL</i>	

NAME:

CLASS:

MATRIC NO:

This question paper consists of 10 printed pages.

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_0	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_0	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$
Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$

Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|--|---|
| 1. $v = u + at$ | 23. $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$ |
| 2. $s = ut + \frac{1}{2} at^2$ | 24. $\omega^2 = \omega_0^2 + 2\alpha\theta$ |
| 3. $v^2 = u^2 + 2as$ | 25. $\theta = \frac{1}{2}(\omega_0 + \omega)t$ |
| 4. $s = \frac{1}{2}(u + v)t$ | 26. $\tau = rF \sin \theta$ |
| 5. $v = u - gt$ | 27. $\frac{Q}{t} = -kA \left(\frac{\Delta T}{L} \right)$ |
| 6. $v^2 = u^2 - 2gs$ | 28. $v_{rms} = \sqrt{\langle v^2 \rangle}$ |
| 7. $s = ut - \frac{1}{2} gt^2$ | 29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$ |
| 8. $p = mv$ | 30. $\Delta U = Q - W$ |
| 9. $J = F\Delta t$ | |
| 10. $J = \Delta p = mv - mu$ | |
| 11. $f_s \leq \mu_s N$ | |
| 12. $f_k = \mu_k N$ | |
| 13. $W = \vec{F} \cdot \vec{s} = Fs \cos \theta$ | |
| 14. $K = \frac{1}{2} mv^2$ | |
| 15. $U = mgh$ | |
| 16. $U_s = \frac{1}{2} kx^2 = \frac{1}{2} Fx$ | |
| 17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$ | |
| 18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$ | |
| 19. $s = r\theta$ | |
| 20. $v = r\omega$ | |
| 21. $a_t = r\alpha$ | |
| 22. $\omega = \omega_0 + \alpha t$ | |

Answer **all** questions.

1. (a) Convert the quantities below to SI units.

- (i) mass of a particle of 8 mg.
 (ii) volume of an iron wire of 50 mm³.

[2 marks]

- (b)

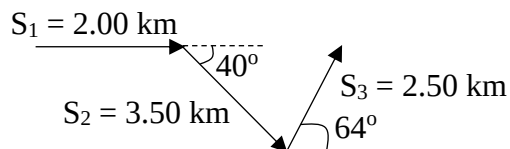


FIGURE 1

A sailor in a small sailboat encounters shifting winds as shown in **FIGURE 1**. She sails 2.00 km east, 3.5 km southeast and 2.50 km northeast. Calculate the magnitude and direction of her resultant journey.

[6 marks]

2. (a)

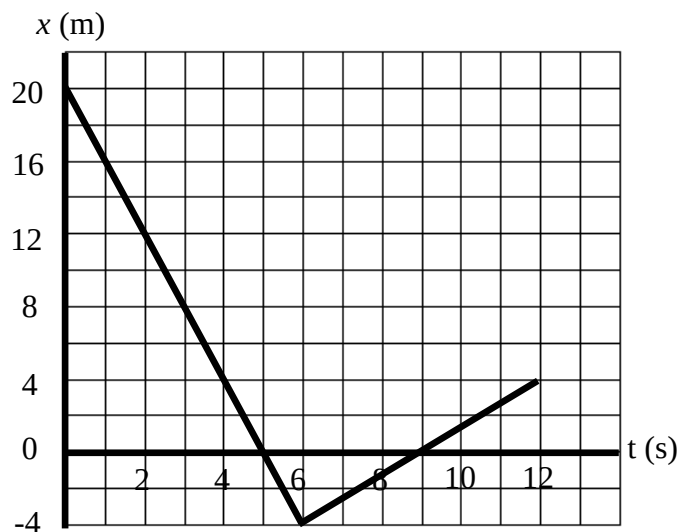


FIGURE 2

FIGURE 2 shows the distance versus time graph of a man walking in the street.

- (i) Sketch a v versus t graph to represent the motion.
 (ii) Calculate his average speed for the whole journey.
 (iii) Calculate his average velocity for the whole journey.

[8 marks]

- (b) A construction worker accidentally drops a brick from the top of a 50 m building.
- (i) What is the velocity of the brick after 1.5 s?
- (ii) How high does the brick fall from the ground after 2.2 s?

[4 marks]

3. (a) A ball of mass 15 g moving at a horizontal speed of 20 m s^{-1} hits a vertical wall. The ball bounces back horizontally with a speed of 10 m s^{-1} after touching the wall for 0.012 s. Calculate the magnitude of force exerted by the ball on the wall.

[2 marks]

- (b) A truck traveling at 50 km h^{-1} strikes a stationary 100 kg car. The vehicles interlock and move off together at the velocity of 49 km h^{-1} . Calculate the mass of the truck.

[2 marks]

- (c)

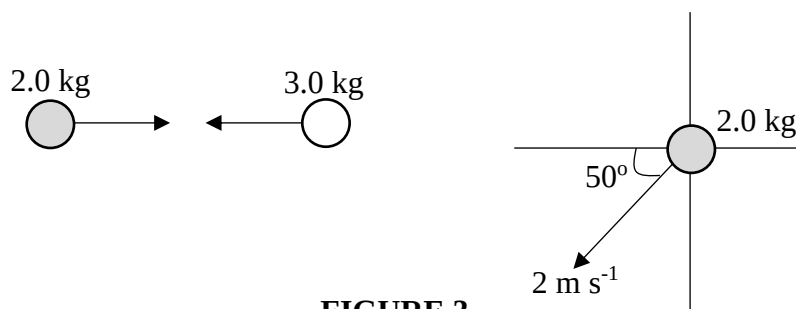


FIGURE 3

A 2.0 kg object with a velocity of 3.5 m s^{-1} to the right strikes a second object of 3.0 kg moving with a velocity of 5 m s^{-1} in the opposite direction as shown in **FIGURE 3**. After an elastic collision, the 2.0 kg is deflected from its original direction through an angle 50° below the negative x-axis and its velocity after the collision is 2.0 m s^{-1} . Calculate the magnitude of the velocity of the 3.0 kg after the collision.

[5 marks]

4. (a) An object with a weight of 150 N is pulled to the right across a rough floor by a force of 90 N. The object moves at a constant speed. Calculate the coefficient of kinetic friction between the object and the rough surface.

[4 marks]

(b)

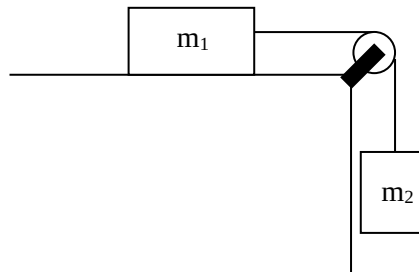


FIGURE 4

FIGURE 4 shows a block of mass $m_1 = 6.0$ kg on a horizontal surface with a coefficient of kinetic friction, $\mu_k = 0.22$ is connected by a string through a pulley to another block mass $m_2 = 8.0$ kg. The system is released from rest.

- (i) Draw the free body diagram of the blocks when they are in motion.
(ii) Calculate the acceleration of the blocks and the tension in the string.

[8 marks]

5. (a) A particle is moving around in a circle of diameter 3.0 m with a constant speed of 2 m s^{-1} . Calculate the

- (i) centripetal acceleration and
(ii) period of the particle.

[5 marks]

- (b) A child twirls his yo-yo about his head rather than using it properly. The yo-yo has a mass of 150 g and is attached to a string 90 cm long.

- (i) If the yo-yo makes a complete revolution each second, what tension must exist in the string?
(ii) If the child increases the speed to 11.31 m s^{-1} , what tension must be in the string?

[4 marks]

6. (a) A bicycle slows down uniformly from 8.40 m s^{-1} to rest over a distance of 115 m. Each wheel and tire have an overall diameter of 68.0 cm. Calculate the
- angular velocity of the wheels at the initial instant, $t = 0 \text{ s}$.
 - total number of revolutions each wheel rotates before coming to rest.
 - angular acceleration of the wheel.

[7 marks]

(b)

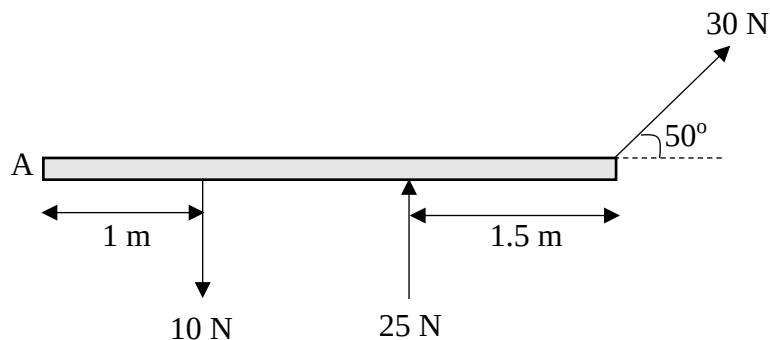


FIGURE 5

A uniform plank of length 4.0 m is exerted by three forces as shown in **FIGURE 5**. If the point **A** is taken as an axis of rotation and the weight of the plank is ignored, calculate

- the torque that rotates in a clockwise and anticlockwise direction.
- the total torques of the system.

[5 marks]

7. (a) The end of an insulated copper rod of radius 1.5 cm and length 30 cm are maintained at 20°C and 60°C . Given the thermal conductivity of the copper rod is $400 \text{ W m}^{-1} \text{ K}^{-1}$. Calculate
- the rate of heat flow through the rod.
 - the temperature at a point 10 cm from the hot end when a steady state is reached.

[4 marks]

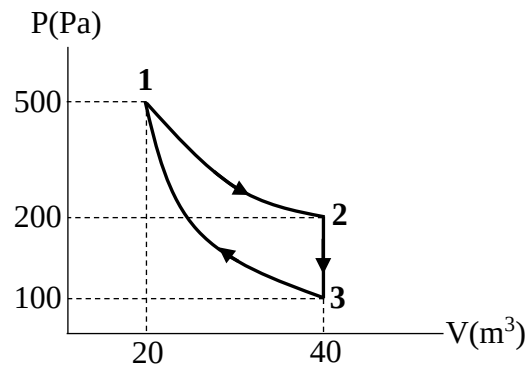
- (b) Five gas molecules chosen at random are found to have speeds of 450 m s^{-1} , 720 m s^{-1} , 560 m s^{-1} , 910 m s^{-1} and 840 m s^{-1} . Calculate
- (i) the mean square speed.
 - (ii) the root mean square speed.

[3 marks]

- (c) (i) When 100 kJ of work is done on a closed system during the process, the internal energy of the system increases by 55.0 kJ. Calculate how much heat is either added or removed from the system.
- (ii) A piston-cylinder system is used to compress 1 kg of gas from 14 m^3 to a volume of 9 m^3 at a constant pressure of 300 Pa. It is found that the internal energy decreases by 8136 J and heat is removed by 9636 J during the process. Calculate the amount of work done for that system and state the type of work done for the system.

[5 marks]

(d)

**FIGURE 6**

Three processes form a thermodynamics cycle as shown in **FIGURE 6** for an ideal gas. Process **1** to **2** occurs at a constant temperature, process **2** to **3** takes place at a constant volume and there is no heat transferred into or out of the system during process **3** to **1**. Specifically, 40 J of heat leaves the system during the process **2** to **3**.

- (i) State the thermodynamics process from **1** to **2**, **2** to **3** and **3** to **1** and specify whether the process involves compression or expansion.
- (ii) Calculate the change in internal energy of the process **2** to **3**.

[6 marks]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT